Library Management System

■ Objective:

To design and implement a Library Management System using Java that demonstrates the key Object-Oriented Programming (OOP) concepts: Classes and Objects, Inheritance, Polymorphism, Encapsulation, Abstract Classes, and Interfaces.

■ Project Description:

A library contains different types of books — such as textbooks, magazines, and novels. Each book has certain common details like title, author, and publication year. However, different types of books may also have unique characteristics (for example, a textbook may have a subject name, a magazine may have an issue number, and a novel may have a genre). Your task is to build a simple Java program that represents this system and shows how OOP principles can be used to organize, extend, and manage data effectively.

■■ Detailed Requirements:

- **1■■ Abstract Class: Book**
  Create an abstract class named Book that will serve as the base class for all types of books. It should contain:
  - Private data members (title, author, yearOfPublication).
  - A constructor to initialize these values.
  - Getters and Setters for each private variable.
  - An abstract method displayDetails().
- **2■■ Subclasses of Book:**
  Create at least two or more subclasses that extend Book, such as TextBook, Magazine, and Novel. Each subclass should:
  - Call the parent class constructor using super().
  - Implement the displayDetails() method.
  - Contain its own additional variable like subject, issueNumber, or genre.
- **3■■ Interface: LibraryOperations**
  Create an interface that declares methods like borrowBook() and returnBook(). Each subclass should implement this interface and provide its own version of these methods.
- **4■■ Encapsulation:**
  Ensure all instance variables are private and accessed through getters and setters.
- **5■■ Polymorphism Demonstration:**
  Use an array or list of Book references and store objects of different subclasses. Call displayDetails() to demonstrate runtime polymorphism.
- **6■■ Main Class: LibraryManagement**
  Create a main class that creates objects, displays their details, and demonstrates interface methods.

■ Concepts to be Demonstrated:

Class & Object – Classes like Book, TextBook, Magazine, Novel, and objects of each type.
Inheritance – Subclasses inherit from the abstract class Book.
Polymorphism – Overriding displayDetails() and using parent class references.
Encapsulation – Private variables with getters and setters.
Abstract Class – Book as the abstract class.

Interface – LibraryOperations defining actions like borrow and return.

■ Expected Output:

The program should display book details (title, author, type, etc.) and borrowing or returning actions. Example:

Book Title: Java Programming
Author: James Gosling
Year: 2020
Subject: Computer Science
This textbook can be borrowed for 15 days.

Book Title: Reader's Digest
Author: John Doe
Year: 2024
Issue Number: 12
This magazine can be borrowed for 7 days.

■■ Submission Guidelines:

- Submit your .java files or zipped folder containing the full project.
- Include proper comments explaining each concept.
- Make sure the program compiles and runs without errors.
- Use meaningful variable and class names.